SMART INDIA HACKATHON PROPOSAL

Tactical Command Trainer for DDIL Environments

Problem Statement ID: 26248
Domain: Defence & Cyber Security

This blueprint outlines a realistic, low-bandwidth, interactive training platform designed to train commanders to operate during **Denied, Degraded, Intermittent, and Limited (DDIL)** communications. It bridges physical electronic warfare realities with cognitive decision science.

01. The Core Issue: Information Starvation & Deception

Modern command platforms assume a continuous, high-speed internet/SATCOM link. However, during peer-conflict, adversaries execute Electronic Warfare (EW) jamming, GPS spoofing, and cyber attacks. This creates three critical failures in command operations:

1. Stale Tactical Maps

Positions freeze or update in bulk every 10–15 minutes. Commanders make decisions based on outdated target coordinates.

2. Partial Telemetry

Packet dropouts strip out key metadata like Friend-or-Foe (IFF) tags or fuel status, leading to misinterpretation.

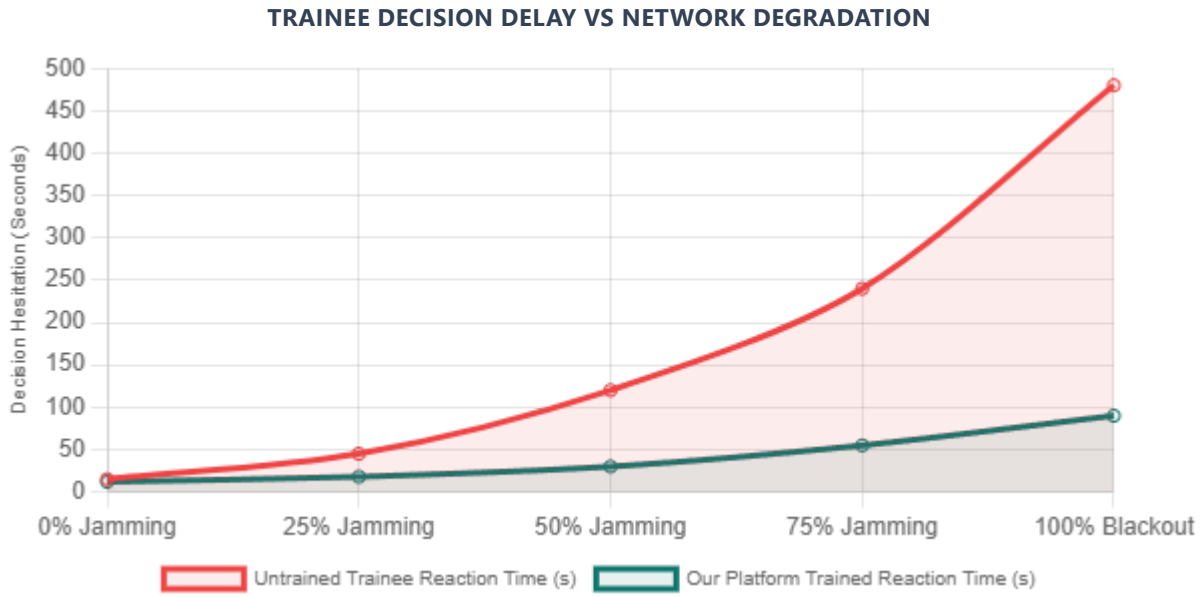
3. Cyber Ghost Tracks

Injected fake C2 messages create false threat tracks, forcing commanders to waste strikes on non-existent targets.

> **The Problem with Current Training:** Military commanders are accustomed to pristine digital displays during peacetime exercises. When faced with degraded displays, they experience *Analysis Paralysis* or make reckless assumptions.

02. Existing Approaches vs. Proposed Solution

Feature / Capability	Traditional C2 Simulators	Our SIH Proposed Platform
Network Modeling	Binary (100% Online or 100% Disconnected)	Dynamic Fading & Packet Burst Loss (Gilbert-Elliott Model)
Uncertainty Display	Shows crisp icons with absolute certainty	Subjective Logic (Belief / Disbelief / Uncertainty circles)
Cognitive Assessment	Evaluates only outcome (Did target get destroyed?)	Separates network latency from decision delay (SAGAT Probes)
Bandwidth Efficiency	Heavy video/graphic streams (Fails on 2G/Radio)	Compressed-DIS / Lightweight JSON telemetry state parser



03. Light Touch: Key Science & Concepts

We use two simple mathematical concepts to power the simulation without overloading system resources:

1. Gilbert-Elliott Network Model

Instead of predicting every radio wave, the channel flips between two dynamic states:

- **State G (Good):** 99% packets pass smoothly.
- **State B (Bad):** 95% packets dropped due to jamming.

Why it matters: It creates realistic "bursts" of network loss rather than simple random drops.

2. Subjective Logic (3-Part Trust Metric)

Instead of giving a simple probability (e.g., "80% chance enemy"), every target track is split into:

```
Opinion = [ Belief + Disbelief + Uncertainty = 1.0 ]
```

Why it matters: If satellite feeds drop, "Uncertainty" rises, signaling to the commander that data is missing, not that the target disappeared.

04. System Architecture & Stack

Frontend Display
Tailwind CSS, HTML5 Canvas, Chart.js for real-time tactical map render & dashboards.

Degradation Engine
Node.js / Python proxy layer applying Gilbert-Elliott packet delays & dropped frames.

Protocol Layer
C2SIM / JSON message formatting for standardized command orders & position reports.

Analytics / AAR
xAPI format event logger tracking decision timestamps vs. ground truth.

05. SIH Hackathon Implementation Roadmap

A realistic execution timeline designed for hackathon milestones and prototype delivery:

PHASE 1

Core C2 Map & Proxy

Build lightweight web tactical display with grid units and basic order buttons.

Deliverable: Working Tactical Map

PHASE 2

Degradation Engine

Integrate Gilbert-Elliott slider controls to introduce live lag, drops, and ghost tracks.

Deliverable: Live EW Simulator

PHASE 3

SAGAT Freeze Probes

Implement unexpected pause probes asking trainees to recall lost status or predict tracks.

Deliverable: Cognitive Tester

PHASE 4

AAR Dashboard

Automated After-Action Review breaking down network lag vs cognitive hesitation.

Deliverable: Final SIH Demo

06. Why This Solution Is Better

- ✓ **Solves Real Defence Need:** Trains commanders to make effective tactical decisions when screens fail, rather than suffering analysis paralysis.
- ✓ **Ultra Lightweight & Portable:** Runs directly in a standard browser over low-bandwidth tactical radios or 2G field networks with zero heavy video dependencies.
- ✓ **Unbiased Cognitive Analytics:** Mathematically separates physical network latency ($\Delta T_{network}$) from human decision hesitation ($\Delta T_{cognitive}$).
- ✓ **Open Defense Standards Compliant:** Uses standard C2SIM and Compressed-DIS data schemas for seamless integration into existing military C4ISR networks.
- ✓ **Realistic Stochastic EW Modeling:** Replaces simplistic binary toggles with Gilbert-Elliott Markov chains that model authentic burst loss and channel fading.
- ✓ **Quantifiable Uncertainty Metrics:** Uses Subjective Logic (Belief / Disbelief / Uncertainty) to explicitly visualize data ignorance rather than hiding it behind misleading binary icons.
- ✓ **Active Cyber Deception Resilience:** Simulates synthetic ghost tracks and altered C2 metadata, actively building critical verification instincts against adversary spoofing.
- ✓ **Objective SA Measurement:** Features automated SAGAT freeze probes to measure situational awareness across perception, comprehension, and projection tiers.
